

D278

Scripting and Programming

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A script that tallies inventory for a hardware store in Maine sets $a = 4$ and $b = 9$, then runs $a = b$ followed by $b = a$. The owner checks the totals every Monday. What are a and b now?

- A. $a = 9, b = 9$
- B. $a = 4, b = 9$
- C. $a = 9, b = 4$
- D. $a = 4, b = 4$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which sequence correctly swaps the values of a and b ?

- A. $a = b; b = a$
- B. $temp = b; a = b; b = temp$
- C. $temp = a; b = a; a = temp$
- D. $temp = a; a = b; b = temp$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A web form used by 300 customers reads two quantities as text, "5" and "3", and the script computes $total = first + second$ without converting them. What is total?

- A. 8
- B. 15
- C. An error, since $+$ is not defined for text
- D. "53"

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A payroll script written by a contractor in 2024 computes $result = 10 - 4 / 2 + 3 * 2$. What is result?

- A. 2
- B. 12
- C. 14
- D. It cannot be known; every language orders these differently.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

With $x = 2$, which expression evaluates to 20?

- A. $x * (3 + 4)$
- B. $x * 4 + 3$
- C. $(x + 3) * 4$
- D. $x + 3 * 4$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A teacher with 28 students posts grades on Fridays. The script is: if score ≥ 60 then grade $\leftarrow$ D; else if score ≥ 80 then grade $\leftarrow$ B; else if score ≥ 90 then grade $\leftarrow$ A. A student scores 85. What grade is assigned?

- A. D
- B. D, then B
- C. B
- D. No grade

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

How should that grading chain be fixed so every score gets the intended grade (A for 90 and up, B for 80–89, D for 60–79)?

- A. Test ≥ 80 , then ≥ 90 , then ≥ 60 .
- B. Change each $\geq$ to $>$.
- C. Test ≥ 90 , then ≥ 80 , then ≥ 60 , in that order.
- D. Add a final else branch that assigns B.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A lab that runs three shifts uses: $i = 1$; while $i < 20$: $i = i * 2$. What is the value of i after the loop ends?

- A. 16
- B. 32
- C. 20
- D. 5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

`i = 10; while i > 0: i = i - 3.` How many times does the loop body run?

- A. 4
- B. 3
- C. About 3.33
- D. -2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A bookstore chain with 30 locations uses function `addTax(price) { total = price * 1.08 }` and then runs `x = addTax(100)`. What is stored in `x`?

- A. 100
- B. 8
- C. 108
- D. Nothing (null / None / undefined)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A program for a museum kiosk has 80 lines and a syntax error on line 50. It is run with an interpreter. What happens?

- A. It produces an executable file with the error inside.
- B. Lines 1–49 run, then it stops with an error at line 50.
- C. It skips line 50 and always finishes.
- D. Nothing runs; the error is reported first.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A driving school with 4 instructors uses: `allowed = (age ≥ 18) OR (hasPermit AND age ≥ 16)`. A student is 17 and `hasPermit` is true. What is `allowed`?

- A. An error; AND and OR can never be mixed
- B. true
- C. false, since OR needs both sides true
- D. false

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A script ~~that tallies inventory for a hardware store in Maine~~ sets $a = 4$ and $b = 9$, then runs $a = b$ followed by $b = a$. ~~The owner checks the totals every Monday.~~ What are a and b now?

Target word: final values **Decides:** After $a = b$, a is 9; then $b = a$ copies 9.

A	Correct	The original 4 was overwritten before it could be copied.
B	Dropped element	Ignores both assignments.
C	Confusion set	Treats it as a swap, which needs a temporary variable.
D	Inverted operation	Runs the second line first.

2. Answer D Concept 1 · transfer

Which sequence correctly swaps the values of a and b ?

Target word: correct swap **Decides:** Save one value before overwriting it.

A	Dropped element	No temporary; both end up equal.
B	Confusion set	Saves the wrong variable; both end up as b .
C	Inverted operation	Copies in the wrong direction; nothing changes in a and b ends equal to a .
D	Correct	temp keeps a 's original value.

3. Answer D Concept 2

A web form ~~used by 300 customers~~ reads two quantities as text, "5" and "3", and the script computes $\text{total} = \text{first} + \text{second}$ without converting them. What is total?

Target word: result of string + **Decides:** Both values are strings.

A	Confusion set	That needs conversion to numbers first.
B	Inverted operation	Multiplies.
C	Adjacent category	Most languages define $+$ as concatenation for strings.
D	Correct	$+$ joins strings.

4. Answer C Concept 3

A payroll script ~~written by a contractor in 2024~~ computes $\text{result} = 10 - 4 / 2 + 3 * 2$. What is result?

Target word: value of the expression **Decides:** Divide and multiply first: $10 - 2 + 6$.

A	Inverted operation	Adds $2 + 6$ before subtracting from 10.
B	Dropped element	Works strictly left to right, ignoring precedence.
C	Correct	$10 - 2 + 6 = 14$.
D	Absolute language	Standard precedence is shared by mainstream languages.

5. Answer C Concept 3 · transfer

With $x = 2$, which expression evaluates to 20?

Target word: pick expression from result **Decides:** Parentheses force the addition first.

A	Confusion set	$2 \times 7 = 14$.
B	Adjacent category	$8 + 3 = 11$.
C	Correct	$5 \times 4 = 20$.
D	Dropped element	Without parentheses: $2 + 12 = 14$.

6. Answer A Concept 4

A teacher ~~with 28 students posts grades on Fridays~~. The script is: if score ≥ 60 then grade $\leftarrow$ D; else if score ≥ 80 then grade $\leftarrow$ B; else if score ≥ 90 then grade $\leftarrow$ A. A student scores 85. What grade is assigned?

Target word: branch that runs **Decides:** $85 \geq 60$ is true first.

A	Correct	The first true branch runs and the rest are skipped.
B	Adjacent category	Describes separate if statements, not else-if.
C	Confusion set	Expects the best match, not the first match.
D	Inverted operation	A branch is true, so one runs.

7. Answer C Concept 4 · transfer

How should that grading chain be fixed so every score gets the intended grade (A for 90 and up, B for 80–89, D for 60–79)?

Target word: correct ordering **Decides:** Check the highest threshold first.

A	Inverted operation	Puts ≥ 80 ahead of ≥ 90 , so 95 would get B.
B	Adjacent category	$85 > 60$ is still true first.
C	Correct	The first true test is now the most specific.
D	Dropped element	85 still matches ≥ 60 first, so the else never runs.

8. Answer B Concept 5

~~A lab that runs three shifts uses:~~ $i = 1$; while $i < 20$: $i = i * 2$. What is the value of i after the loop ends?

Target word: final value **Decides:** $1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 16 \rightarrow 32$; 32 fails the test.

A	Dropped element	Stops at the last value below 20 without the final pass.
B	Correct	$16 < 20$ runs once more, giving 32.
C	Wrong stem number	The bound, not a value i reaches.
D	Confusion set	That is the number of passes.

9. Answer A Concept 5 · transfer

$i = 10$; while $i > 0$: $i = i - 3$. How many times does the loop body run?

Target word: pass count **Decides:** $10 \rightarrow 7 \rightarrow 4 \rightarrow 1 \rightarrow -2$.

A	Correct	Runs at 10, 7, 4, and 1.
B	Dropped element	Misses the pass when i is 1.
C	Adjacent category	Divides 10 by 3; passes are whole.
D	Confusion set	That is the final value.

10. Answer D Concept 6

~~A bookstore chain with 30 locations uses~~ function `addTax(price) { total = price * 1.08 }` and then runs `x = addTax(100)`. What is stored in `x`?

Target word: value returned **Decides:** No return statement.

A	Wrong stem number	The argument, not a return value.
B	Adjacent category	The tax amount, which is neither computed alone nor returned.
C	Confusion set	total is computed but never returned.
D	Correct	Without return, no value comes back.

11. Answer B Concept 7

~~A program for a museum kiosk has 80 lines and~~ a syntax error on line 50. It is run with an interpreter. What happens?

Target word: interpreted behavior **Decides:** Interpreters run statements in order.

A	Adjacent category	Interpreters do not create executables.
B	Correct	The error is found only when reached.
C	Absolute language	Errors stop execution.
D	Confusion set	That is compiled behavior.

12. Answer B Concept 8

~~A driving school with 4 instructors uses:~~ `allowed = (age ≥ 18) OR (hasPermit AND age ≥ 16)`. A student is 17 and `hasPermit` is true. What is `allowed`?

Target word: value of compound condition **Decides:** `false OR (true AND true)`.

A	Absolute language	Parentheses make mixing clear and legal.
B	Correct	The second part is true, so OR is true.
C	Confusion set	That is AND.
D	Dropped element	Looks only at <code>age ≥ 18</code> .

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Assignment is sequential	$x = y$ copies y 's current value into x and overwrites x ; statements run top to bottom.	Reading $a = b$; $b = a$ as a swap.	Two or three assignments between the same variables.
2	Data types change what operators do	$+$ adds numbers but joins (concatenates) strings; $"5" + "3"$ is $"53"$.	Assuming text that looks numeric behaves like a number.	Values read from input or wrapped in quotes.
3	Operator precedence	Multiplication and division happen before addition and subtraction; equal levels go left to right; parentheses override.	Evaluating strictly left to right.	A mixed expression without parentheses.
4	If / else-if order	In an if-else-if chain, only the first true branch runs.	Expecting the best-matching branch or all true branches to run.	Overlapping conditions like ≥ 60 and ≥ 80 .
5	Loop tracing	Check the condition before each pass; the body runs only while it is true, and the variable keeps its last updated value.	Stopping one pass early or confusing the pass count with the final value.	A while loop with multiply or subtract updates.
6	Functions return values explicitly	A function sends a value back only through return; without it, the caller gets nothing (null, None, or undefined).	Assuming the last computed value comes back automatically.	A function that computes but never says return.
7	Compiled vs. interpreted	A compiler translates the whole program before any of it runs; an interpreter translates and runs one statement at a time.	Thinking an interpreted program refuses to start if any line has an error.	An error partway through a program and what runs first.
8	Boolean logic	AND is true only if both sides are true; OR is true if either side is true; parentheses group first.	Treating OR like AND.	A compound condition joined by AND and OR.

Course scope (public description): Foundations - Scripting and Programming - Foundations introduces programming basics such as variables, data types, flow control, and design concepts. The course is language-agnostic in nature, ending in a survey of languages, and introduces the distinction between interpreted and compiled languages. Learners will gain skills in identifying scripts for computer program requirements and in using fundamental programming elements as part of common computer programming tasks. Learners will also gain an understanding of the logic and outcome of simple algorithms.